

Development

Age	Primitive Newborn Reflexes	Gross Motor	Fine Motor	Cognitive	Language	Social	Red Flags *Autism	
						Erikson's		
Newborn		Head lag	Blurry vision Fix and follow a near fact or light	Piaget's: Sensorimotor (0-2 years) -Primary circular (own body) -Secondary circular (imitating and repeating pleasurable behavior) -Tertiary circular (explores world, engages in novel or unfamiliar behavior)		Endogenous Smile	Trust vs. Mistrust (0-18 months)→ Basic trust in the world.	
6 weeks		Lift head to 45 degree angle while lying prone	Turn head (90 degrees) to follow object					
2 months					Coo	Social smile		
4 months	Sucking reflex stops Rooting reflex stops	Holds head upright in a sitting position	Hand regard Reaches for objects					
6 months	Palmar grasp reflex stops Moro (Startled) reflex stops	Sits unsupported Rolls over from tummy to back	Wide hand grip Transfers object between hands		Babbles	Stranger anxiety (6 months-12 months)	Cannot hold head up unsupported *Not reaching for objects, *No social smile	
7 months				Forming a schema (schema, assimilation, accommodation)				
9 months		Crawls Pulls to stand	Immature pincer grasp (3 fingers)	Object permanence (transition into preoperational Piaget's stage)	Mama and Dada	Separation anxiety (9 months-2 years)	Cannot sit unsupported *Limited eye contact	
12 months		Walks unassisted	Mature pincer grasp (thumb and index fingers) Points with index finger		10 words 1 word phrases		No pincer grasp	
18 months			Scribbles Hand preference Build 3-block tower			Solitary play (<18 months)	No speech, No attempts to walk *No interest in social interaction	
2 years	Babinski Reflex stops	Walks up and down stairs Jumps Throws and kicks ball	Uses utensils Build 6-block towers	Naming body parts	200 words 2 word phrases Telegraphic sentences	Parallel play (18 months-2 years)	Autonomy vs. Shame (18 months-3 years)→ Sense of independence and self control	Cannot walk independently
3 years		Rides tricycle Walks down stairs with alternating feet	Brushes teeth Builds 9-block towers Draws circle	Piaget: Preoperational Stage (Symbolic thinking, egocentric, no law of conservation)2-7 yrs. Death is reversible	300+ Words 3-word sentences Knows first name	Cooperative play (>2 years) Gender identity Complex emotions		
4 years		Hop on one foot Voluntary bowel control	Dress themselves (uses buttons and zips) 4 years → draws cross 4.5 years → Draws rectangle	Piaget: Preoperational Stage (Symbolic thinking, egocentric, no law of conservation) - Play - words, smile is not evil - - Can't appreciate other perspectives - has spent most of life on things of external value of others	4-word sentences Can talk about day Names colors	Imaginary play (fears begin) Romantic feelings	Initiative vs. Guilt (4-6 years)→ Sense of purpose and initiative	
5 years		Skip Walk backward Voluntary bladder control	Tie shoelaces Knows left vs. right Print letters Draw square	Piaget: Preoperational Stage (Symbolic thinking, egocentric, no law of conservation)	5-word sentences Count to 10 Stories	Imaginary play (fears begin) Romantic feelings		
6 years			Handedness established left vs. right Draw triangle	Piaget: Preoperational Stage (Symbolic thinking, egocentric, no law of conservation) Understand they will die (fear death) ghosts		Uses words more than actions to express emotions, Social display rules about emotions	Industry vs. Inferiority (6-12 years)→ Sense of competence and mastery	
7-12 years			7 years→ Draw diamond	Piaget: Concrete Operational7-12 Stage (Logical thinking, law of conservation established) 10-12 years→ Understand death (interested in biology)		Learning social rules Team player Other's perspective		
12-21 years	Libido system matures before. Prepubertal desire increased risk-taking and peer impulse control	Puberty (girls- 11 to 12, boys- 14 years) Sleep phase delay		Piaget: Formal operational (12- adulthood) (scientific methods, hypothetical thinking) Death is irreversible		Independence body image, peer relationships, love of identity, love of family Risky behaviors	Identity vs. Role Confusion (12-20 years)→ Sense of identity and purpose	
18-29 years (Emerging Adulthood)		Peak physical health and cognitive skills Peak risk for mental health disorders		Peak physical health and cognitive skills		Exploration (education, careers, relationships) Instability (changing residence, jobs, relationships)	Intimacy vs. Isolation (20-40 years)→ Ability to form close and meaningful relationships Generativity vs. Stagnation (40-60 years)→ Feeling of contributing to the world	
30-60 years (Middle Adulthood)		Sleep phase advance Increased risk for chronic diseases (Hypertension,		White matter increase until ~40 years Peak fluid intelligence		Single vs. Partnership vs. Married vs. Divorced;		

		osteoporosis, osteoarthritis) Changes in sensory acuity (presbyopia) Menopause in women (short time, noticeable symptoms) Andropause in men (long time, little to no symptoms)				Child free (by choice) vs. Childless (not by choice) vs. Parenthood	
≥ 60 years (Late Adulthood)		Declining health Risk for polypharmacy Frailty syndrome		CSF volume sharply increases ~60 years (visible brain atrophy and enlarged ventricles) Fluid intelligence decline (crystallized intelligence fine)		Retirement Boomerang children Shrinking social network (Loneliness and Social isolation) Loss	Integrity vs. Despair (≥60 years) → Sense of wholeness and satisfaction with life

Effective Communication Skills

● (1) Open-Ended Skills

○ Non-Focusing

■ Silence

- Encourages deeper expression from the patient without interruption

■ Non-verbal communication (listening behavior)

- Body language, facial expressions, and gestures show attentiveness and understanding

■ Neutral utterances and continuers

○ Focusing

■ Reflection (echoing)

■ Open ended questions

■ Summarizing

● (2) Emotion Speaking Skills

○ Direct

■ Asking emotion focused open-ended questions

■ Encouraging expression of emotions

○ Indirect

■ Eliciting and exploring emotions without directly asking about them

● (3) Empathy Skills (NURS)

○ Naming

■ Repeat the expressed emotion name

○ Understanding

■ Deepen your understanding of emotion

○ Respecting

■ Appreciated and acknowledge the patient's response is valid

○ Supporting

■ Offer support

Stages of the Behavioral Change Model

● Mnemonic → PCP are mentally retarded

○ Precontemplation

○ Contemplation

○ Preparation

○ Action

○ Maintenance

○ Relapse

- **(1) Precontemplation:** Person is NOT yet considering a change (may be unaware, resistant, or previously failed)
 - Motivational techniques:
 - Build rapport and establish trust→ Use open ended questions to explore views→ Gently raise awareness about risks and provide non-judgemental feedback
- **(2) Contemplation:** Person is aware of problem and is CONSIDERING change but is still AMBIVALENT (weighing the pros and cons)
 - Motivational techniques:
 - Help explore pros and cons
 - Help explore barriers
 - Encourage self reflection and strengthen self efficacy of patient
 - Focus on past successes
- **(3) Preparation:** Person intends to make the change in the near future; person is developing action of plan; person has developed action plan but has not yet implemented
 - Motivational techniques:
 - Assist with goal setting
 - Develop specific action plan
 - Reinforce self efficacy
 - Provide support and encourage their belief
- **(4) Action:** Person actively makes change but it has been ≤ 6 months
 - Motivational techniques:
 - Provide encouragement (promote confidence and self efficacy)
 - Reinforce their success
 - Highlight benefits they are experiencing
 - Help identify high risk situations
 - Discuss coping strategies
- **(5) Maintenance:** Person sustained behavioral change for > 6 months
 - Motivational techniques:
 - Celebrate successes
 - Reinforce confidence in ability to maintain change
 - Encourage use of support systems
- **(6) Relapse:** Return to the previous behavior (can occur at any stage)
 - Motivational techniques:
 - Normalize relapse as part of the process
 - Help understand what led to relapse and discuss lessons learned
 - Reassess motivation and barriers
 - Adjust plan as needed
 - Reinforce importance of continuing the change process

Mental Status Examination (MSE)

- Mnemonic: **ASEPTIC**
 - Appearance and behavior
 - Speech
 - Emotion (mood and affect)
 - Perception
 - Thought content and process
 - Insight and judgment
 - Cognition

● (1) Orientation:

- Patient is oriented 3x→ Person, place, and time
- Patient is oriented 4x→ Person, place, time, and purpose

● (2) Appearance:

- Alertness→ Drowsy, vigilant, lethargic, stuporous, asleep, comatose, confused, fluctuating
- Body Weight→ Normal, overweight, underweight
- Grooming→ Unshaven, lack of bathing, bags under eyes
- Dress→ Appropriate, proactive, meticulous, odd

● (3) Motor Behavior:

- Posture→ Slumped, upright, tense, closed off, withdrawn
- Movements→ Hypoactive (slowed), hyperactive, facial droop, weakness, asymmetries
- Involuntary Movements→ Tremor at rest, ticks, dystonias, etc.

■ **Abnormal Motor Behavior Descriptions:**

- Dystonia: Sustained muscle contractions that can cause twisting (torticollis) or abnormal postures
- Tardive Dyskinesia: Involuntary, repetitive movements of the face, mouth, and tongue (lip smacking, tongue thrusting, chewing, grimacing) as a result of long term use of antipsychotic medications
- Tics: Sudden, recurrent, non-rhythmic actions which may be vocal and/or motor
 - Motor Tics: Brief and purposeless movements that usually involve the face and neck
 - Vocal Tics: Sudden, repetitive involuntary sounds (brief grunts, coughing sounds, howling or barking like noises, coprolalia)
- Stereotypic Movement: Involuntary, repetitive, rhythmic movements (rocking, swaying, hand flapping, head banging, complex motor rituals)
- Nervous Habits→ Nail biting, drumming table, teeth grinding, thumb sucking, picking at skin
- Odd Mannerisms→ Stylized movements, stereotypic movements

● (4) Emotion

- (4a) Mood: SUBJECTIVE assessment of emotional state AS REPORTED BY THE PATIENT

■ **Abnormal Mood and Affect Behavior Descriptions:**

- Depressed: Sad, mournful
- Detached: Inability to fully engage with their own feelings or feelings of others (inability to be empathetic)
- Euthymic: Normal, tranquil mood
- Euphoric: Positive, elated mood
- Irritable: Easily annoyed or made angry (agitated)
- Labile: Sudden and frequent changes in expressed emotions
- (4b) Affect: OBJECTIVE assessment of the patient's OBSERVED emotional state

■ **Abnormal Affect Behaviors Descriptions ONLY:**

- Blunted: DECREASED facial and vocal expression and DECREASED expressive gestures in reaction to emotion provoking stimuli
- Flat: NO emotional responses to situations and events, NO facial and vocal expressions, NO expressive gestures in reaction to emotion provoking stimuli

● (5) Speech and Language:

- Speech→ Amount, rate, tone, volume, fluency, articulation, and spontaneity
- Language→ Language comprehension, language expression
- **Abnormal Speech and Language Behavior Descriptions:**
 - Dysfluent: Repetitions (of sounds, syllables, or words) or prolongations (sounds) in speech
 - Dysprosodia: Unusual changes in pitch, intonation, stresses, phrasing, and rhythm in speech
 - Echolalia: Automatic repetition of somebody else's words

- Paraphasia: Speech that contains word retrieval errors
 - Semantic Paraphasia: Incorrect retrieval in words that are similar in MEANING of intended word
 - Phonemic Paraphasia: Incorrect retrieval in words that are similar in SOUND to intended word
 - Neologistic Paraphasia: Incorrect retrieval in words that is a new word and NOT SIMILAR in sound or meaning to intended word
- Poverty of Speech: Poorly elaborated speech
- Pressured: Abundant, non-stop speech (difficult to interrupt)
- Telegraphic: Omission of connecting words, prepositions, and modifiers when speaking
- (7) **Thought Process/ Form**: Examination of flow/logic of thinking characterized by expression of ideas that are evidenced by speech
 - Sequence of thoughts, logical connections, ability to provide specific information and details
 - Normal Thought Process/ Form: Logical, Goal directed
 - **Abnormal Thought Process/ Form Behavior Descriptions**:
 - Circumstantial: Winding/indirect speech; patient gives unnecessary details before ARRIVING TO THE MAIN POINT
 - Tangential: Poor topic adherence; answer to question that veers off from topic (1-2 times) and DOES NOT return to original topic but there is a logical connection between responses
 - Flight of Ideas: Rapidly jumping from topic to topic (3+ times) without completing each train of thought; often expressed through rapid, pressured speech
 - Derailed: NO logical connection between ideas and loss of an overall sense of goal directedness; words make sentences but the sentences do not make logical sense
 - Clanging: Association of words or ideas based on similar SOUND rather than meaning
 - Neologistic: Words that are created by the patient and do not have meaning to others
 - Preservative: Repetitive thought process in which the patient goes back to the same topic and cannot move past it, regardless of the topic or question; LOSS OF GOAL DIRECTEDNESS
- (8) **Thought Content**: WHAT the patient is thinking about
 - Ideas, beliefs, preoccupations, obsessions
 - **Abnormal Thought Content Behavior Descriptions**:
 - Delusional: Fixed, false beliefs that is at odds with reality (but person holds belief with complete certainty)
 - Mood Congruent: Belief results in positive emotions
 - Mood Incongruent: Beliefs result in negative emotions
 - Bizarre: Beliefs are highly implausible and/or impossible
 - Non-Bizarre: Beliefs are plausible
 - Ruminative: Preoccupation with DISTRESSING memories, thoughts, or worries with the hopes of GAINING NEW INSIGHT; CONTENT of one's thoughts (vs. process of one's thinking in preservative)
 - Obsessional: Intrusive and unwanted persistently recurring thoughts, urges, or images that results in MARKED ANXIETY OR DISTRESS
- (9) **Perception**:
 - **Abnormal Perception Behavior Descriptions**:
 - Hallucinations: Sensory experiences that occur WITHOUT EXTERNAL STIMULATION
 - Auditory, visual, tactile, olfactory, gustatory
 - Illusions: Distorted perception of EXISTING sensory stimuli
- (10) **Insight/ Judgement**:
 - Insight→ Awareness and understanding of the problem
 - Judgment→ Ability to organize and manipulate information to make appropriate decisions and regulate behavior
- (11) **Cognition**:
 - Attenuation, Concentration, Distractibility→ Spell “world” backward, serial 7s, serial 3s
 - Attenuation: Patient's ability to notice, process, and respond to specific stimuli or tasks in the environment
 - Concentration: Ability to sustain mental effort on a task or thought over a period of time

- **Distractionability:** How easily the patient's focus is pulled away from the task at hand by internal thoughts or external stimuli
- **Memory**→ Describe the where and when of important life episodes, name the past 3 presidents of the USA, give 3 words and ask them to recall immediately and after 5 minutes
- **Intellectual**→ Ask how 2 words are similar, ask what proverb means
- **(12) Risk Assessment:** Assess risk of harm to or by the patient (suicide vs. homicide)
 - **Suicidality Predictive Factors:**
 - Male→ more likely to COMPLETE suicide due to lethality
 - Female→ More likely to ATTEMPT suicide
 - Older age (≥ 75)→ Increased COMPLETION rate of suicide
 - Prior attempts
 - Mental health conditions
 - Access to lethal means
 - Substance abuse or trauma history
 - Stressful life events
 - Recent preparations for death
 - **Homicide Predictive Factors:**
 - Male sex
 - History of violence
 - Access to firearms (lethal means)
 - Substance abuse
 - Acute changes in mental status
 - Family history
 - Traumatic brain injury (frontal, disinhibited)
 - Recent stressors
 - **Suicide and Homicide Protective Factors**
 - Positive social support
 - Sense of responsibility to family
 - Children at home, pregnancy
 - Spirituality
 - Life satisfaction
 - Positive coping skills
 - Positive therapeutic relationship
 - **Treatment:** Involuntary hospitalization (without consent) in case of patient being a danger to themselves or to others

Learning Theory (Classical and Operant Conditioning)

- **Classical Conditioning**
 - *Before Learning:* Unconditioned stimulus→ Innate unconditioned response
 - *Learning Phase:* Unconditioned stimulus + Neural stimulus→ Unconditioned Response
 - *After Learning:* Conditioned stimulus→ Learned conditioned response
 - Neutral stimulus becomes conditioned stimulus
 - Unconditioned response becomes conditioned response
- **Classical Extinction**
 - Repeated exposure of conditioned stimulus without conditioned response→ Conditioned stimulus returns to being a neutral stimulus that is unpaired with the unconditioned response
 - *Spontaneous Recovery:* Re-emergence of conditioned response after classical extinction occurs
- **Operant Conditioning**

- Association with behavior (desired or undesired) and consequence (reinforcement or punishment)
 - If behavior is strengthened
 - The behavior has been reinforced
 - The consequence is reinforcing
 - If the behavior is weakened
 - The behavior has been punished
 - The consequence is punishing
- **Reinforcement:** Strengthening of desired behavior
 - **Positive Reinforcement:** Desired behavior→ Pleasurable/desirable stimulus is APPLIED→ Behavior will be repeated
 - **Negative Reinforcement:** Desired behavior→ Adverse/undesirable stimulus is REMOVED→ Behavior will be repeated
- **Punishment:** Weakening of undesired behavior
 - **Positive Punishment:** Undesired behavior→ Adverse/undesirable stimulus is APPLIED→ Behavior will not be repeated
 - **Negative Punishment:** Undesired behavior→ Pleasurable/desirable stimulus is REMOVED→ Behavior will not be repeated
- **Operant Extinction**
 - Applied/removed reinforcement/ punishment no occurs when behavior occurs
 - **Extinction Burst:** Initial escalation of behavior during operant extinction
 - Do not confuse with **Punishment**- Behavior occurs and a DIFFERENT consequence is applied
- **Operant Schedule**
 - Rules that determine how often targeted behavior is reinforced or punished
 - **Continuous Schedule:** RF/P occurs at EVERY occurrence of targeted behavior→ Contingency between behavior and consequence is clearer→ Faster LEARNING
 - **Intermittent Schedule:** Any schedule that is not continuous, results in slower extinction of targeted behavior
 - **Ratio Schedule (Response Driven):** RF/P occurs after a certain number of responses occur
 - Fixed Ratio→ RF/P occurs after predictable number of responses
 - Variable Ratio→ RF/P occurs after unpredictable number of responses
 - **Interval Schedule (Calendar Driven):** RF/P occurs at a specific date or time
 - Fixed Interval→ RF/P occurs after predictable date or time
 - Variable Interval→ RF/P occurs after unpredictable date or time

Psychotherapies

- (1) Behavioral Therapy
 - Based on **Behavioral Theory** (Classical and Operant Conditioning)
 - **Exposure Techniques:**
 - **Graded Exposure**
 - Pacing of exposure→ Gradual
 - Patient must endure anxiety (no safety behaviors)
 - Learning Principle→ Classical Extinction
 - **Exposure and Response Prevention (ERP)** (part of graded exposure technique for OCD)
 - Pacing of exposure→ Gradual
 - Patient must endure anxiety (no safety behaviors)
 - Learning Principle:
 - Classical extinction (exposure component)
 - Operant conditioning (response prevention component)
 - **Flooding**
 - Pacing of exposure→ **Non-gradual**
 - Patient must endure the anxiety (no safety behaviors)

- Learning Principle→ Classical extinction
- **Systematic Desensitization**
 - Pacing of exposure→ Gradual
 - Patient must actively relax (safety behavior)
 - Learning Principle→ Counterconditioning
- **Aversion Therapy**: Weakening the undesirable behavior by associating undesirable behavior with unpleasant responses
 - **Covert Sensitization**: If unpleasant experience is imaginal
 - Learning Principle→ Counterconditioning
- **Biofeedback**: Increasing the control of physiological functions through tracking of bodily feedback and translating it into a signal (auditory or visual display) to help a person to modify their physiological functioning when direct feedback is no longer provided
 - **Desired physiology** is reinforced through *desirable signal* (Positive Reinforcement)
 - Undesired physiology is replaced by Desired behavior→ Desired stimulus applied→ Behavior is likely to be repeated
 - **Undesired physiology** is punished through an *undesirable signal* (Positive Punishment)
 - Undesired behavior→ Undesired stimulus applied→ Behavior is LESS likely to be repeated
 - Learning Principle→ Operant Conditioning
- **Token Economies**: Increase desired behaviors through systematic reinforcement of desired behaviors by giving tokens that are exchanges for reward (positive reinforcement)
 - Sometimes token economies also include REMOVING tokens to decrease undesired behaviors (negative punishment)
 - Risk involved→ Only use negative punishment in extreme display of undesired behaviors
 - Learning Principle→ Operant Conditioning (Positive Reinforcement)
- **Behavioral Modeling**: Learning new skills or ways of responding to situations by observing a desired behavior of a model and then practices the behavior (model can be live or video based)
 - Learning Principle→ Vicarious Conditioning
- (2) **Cognitive Therapy**
 - Based on **Cognitive Theory**: Our thoughts and interpretations affect our emotions and behavior
 - Maladaptive emotions and behaviors result from cognitive distortions
- (3) **Cognitive Behavioral Therapy**
 - Based on both **Cognitive and Behavioral Theories**
- (4) **Psychodynamic (Psychoanalytic) Therapy**
 - Based on **Psychodynamic (Psychoanalytic) Theory**
- (5) **Family Therapy**
 - Based on **Family Systems Theory**